Kadanes Algorithm

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The problem

- Maximum sum of a non-empty contiguous subarray.

Intuition

- The best subarray ending here is either this element alone or it extends the best ending at the previous index.

Approach

- 1. Running best ending here
- 2. Max of element or element plus running
- 3. Track the global max

Java solution

```java
public class KadanesAlgorithm {
    public static int kadanesAlgorithm(int[] a) {
        int cur = a[0], best = a[0];
        for (int i = 1; i < a.length; i++) { cur = Math.max(a[i], cur + a[i]); best = Math.max(best, cur); }
        return best;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- The best subarray ending here is either this element alone or it extends the best ending at the previous index.
- Complexity: O(n)